Industry Served: ***Workforce & Operations***

Pulse Security Management System: A Centralised Platform for Customer, Site, Job, Asset and Project Operations

Replacing Disconnected Tools with a Single Operational Platform for Security Teams

INTEGRATED SECURITY | PULSE Security Management

One Platform for the Full Security Operation

Customers
Sites
Jobs
Equipment
Assets
Parts
Compliance
Projects
CUSTOMERS
SITES
ASSETS
COMPLIANCE

Security installation and service companies manage a high volume of operational complexity: customers spread across many sites, engineers executing jobs in the field, equipment libraries and parts catalogues to maintain, assets to track through their full installed lifecycle, compliance documents to generate, and project pipelines to manage from initial enquiry through to certified completion. Across most operations of this kind, this work is distributed across disconnected tools: spreadsheets, email threads, separate job management systems and manual record-keeping, with no single source of truth for what is happening across the business.

Pulse is a purpose-built security operations management platform developed to solve that problem. It brings customers, sites, jobs, equipment, parts, assets, projects, compliance and team management into a single connected system, with role-based access control, clearance management for sensitive customers, and a cost estimation engine that handles the Irish VAT rules applicable to security installation work.

Project Challenges and Needs

1

Fragmented Operations

Customers, sites, jobs, equipment, parts and compliance documents were managed across separate tools with no connection between them. The platform needed to unify the full operational workflow into one place, eliminating duplication and giving every team member a consistent view of the business.

2

Complex Job Costing

Security installation jobs require accurate cost estimation across labour, parts, callout fees and VAT, with three distinct Irish VAT calculation rules depending on the ratio of parts to labour. The system needed to handle this automatically, with configurable rates and callout fee bands to support international use.

3

Classified Customer Access

Certain customers and their associated sites, jobs and data required restricted visibility, accessible only to engineers with explicit clearance. The platform needed a clearance management system with request and approval workflows, and the ability to mark entire customer profiles as classified at the point of creation.

4

Asset Lifecycle Management

Installed equipment needed to be tracked through its full lifecycle, from job card to asset record, with multi-warranty support (parts and labour at different periods and from different manufacturers), service log history, and the ability to convert an installed piece of equipment into a tracked asset directly from the job.

5

Stage-Gated Project Delivery

Larger security deployments required structured project management from enquiry through to certified completion. Each stage (survey, specification, preparation, deployment, certification) needed its own task lists, approval requirements and notes, with future stages locked until prerequisites were met.

6

Field Engineer Workflow

Engineers working on-site needed to complete tasks, log notes in text, photo, video and audio formats, record site visits with arrival and departure times, book tools, and manage job alterations, all within the same system that management used to plan and schedule their work.

Our Solution

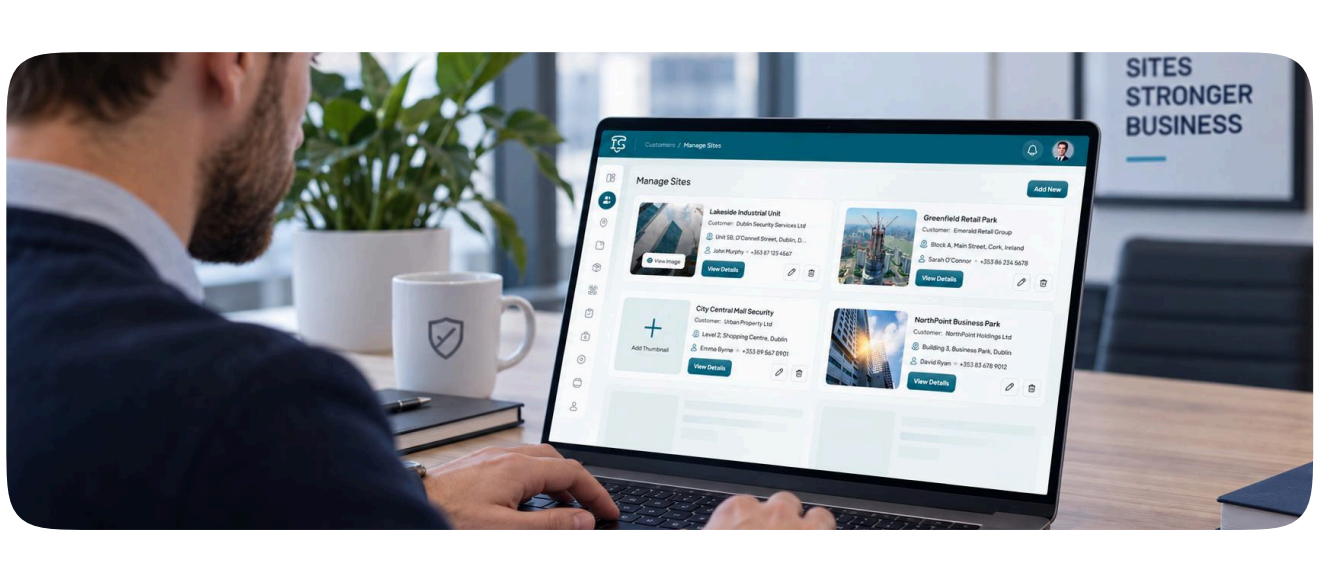
We built a comprehensive security operations management platform, delivering:

- Customer and site management with geographic map view, starred records, account manager linking, and classification toggles for restricted visibility
- Job management with task tracking, completion logging, parts and equipment linking, scheduling, engineer assignment, & multi-band callout fee support
- Automated cost estimation engine handling three Irish VAT calculation methods across labour-only, mixed, and parts-heavy job types, with configurable rates
- Equipment library with parts composition, upgrade paths, licence options, install time with difficulty multipliers, and full margin and cost breakdown
- Parts catalogue with variants, customisations, supplier linkage, per-customer-type discount logic, and Excel import and export
- Asset creation from job cards with multi-warranty support across parts and labour, service log history, media and document storage
- Stage-gated project lifecycle from enquiry through to customer sign-off, with approval requirements at pricing, certification and completion stages
- Compliance document generation from predefined templates, auto-filled with job, site and asset data, downloadable as PDF
- Stage-gated project lifecycle from enquiry through to customer sign-off, with approval requirements at pricing, certification and completion stages
- Compliance document generation from predefined templates, auto-filled with job, site and asset data, downloadable as PDF
- Clearance management for classified customers, with request and approval workflows and role-gated data visibility
- Role-based access control across Admin, Engineer, Solution Engineer and Senior Management, with 2FA via Google or Microsoft Authenticator
- Full audit logging of all key user actions across every module for security and traceability

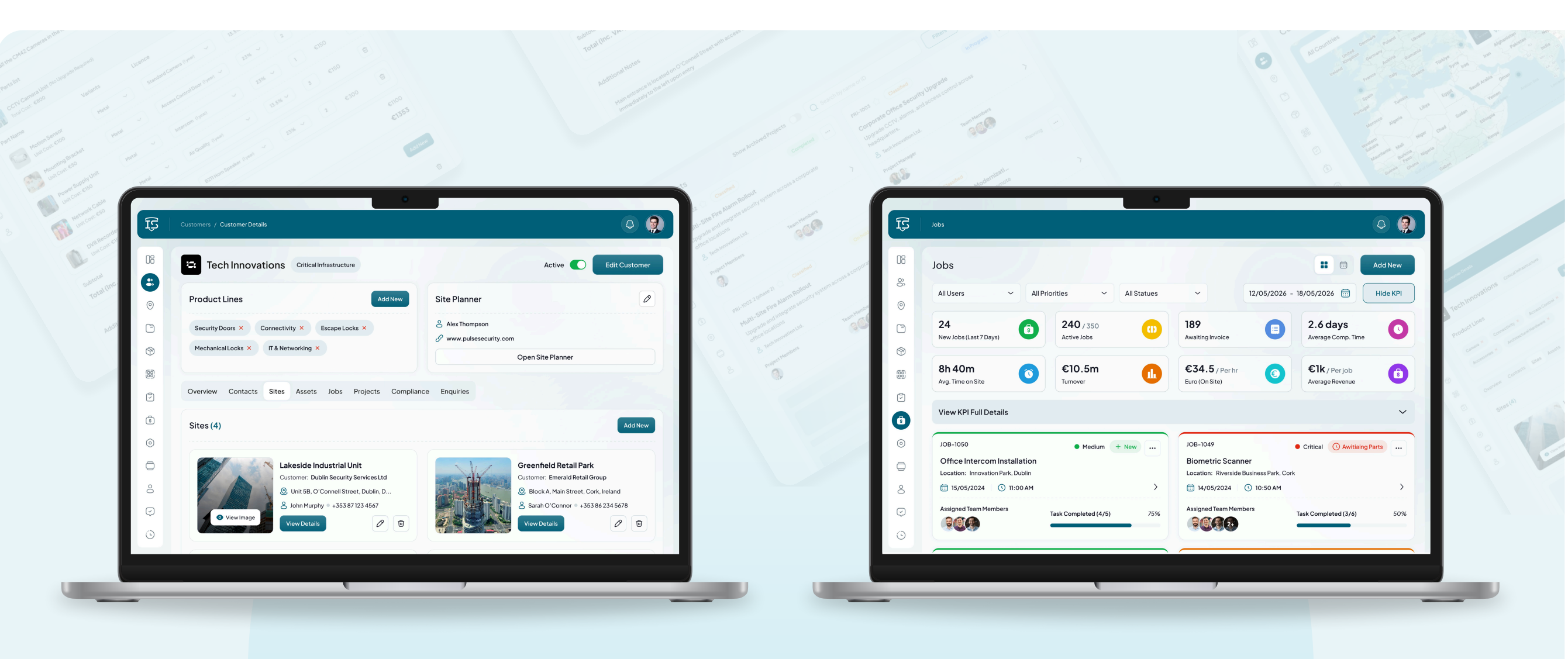
Interfaces We Developed

Web Management Panel:

The primary operational interface for all roles across the business:



- ✓ Dashboard with tile-based module navigation, quick-action buttons and notification indicators
- ✓ Sites module: global and per-customer site views with media uploads, floor plans, walkthrough videos and compliance storage
- ✓ Equipment module: library management with parts, upgrades, licences, pricing configuration and difficulty multipliers
- ✓ Assets module: installed asset tracking with multi-warranty handling, service logs and media storage
- ✓ Compliance: template-based document generation, PDF export and generated document management
- ✓ Customer module: full lifecycle from creation to classified management, with site, job and asset linkage
- ✓ Jobs module: creation, scheduling, task management, cost estimation, site visits, tool bookings, alterations and activity logging
- ✓ Parts module: catalogue management with variants, supplier linkage, discount logic, suggestion workflows and Excel import/export
- ✓ Project management: stage-gated project lifecycle with team assignment, approval workflows and stage-level task tracking
- ✓ Users, audit logs, clearance management, company profile and system configuration



Key Features and Capabilities

Job Management and Cost Estimation

Jobs are created with tasks, linked parts and equipment, scheduling, and engineer assignment. The cost estimation engine automatically calculates labour, callout fees (across three configurable location bands) and parts costs, applying the correct Irish VAT rate based on the proportion of parts to labour, with the ability to override rates for international use.

Equipment and Parts Library

Equipment is defined as complete security solutions, each composed of parts with variants, upgrade options and licence configurations. Pricing flows from component-level cost through to installed retail price, with difficulty multipliers adjusting install time for specialist conditions. Parts carry supplier linkage, percustomer-type discount logic and full variant support.

Asset Lifecycle Tracking

Installed equipment is converted into asset records directly from the job card, with all known data pre-filled. Each asset supports multiple concurrent warranties (parts and labour at different periods and from different manufacturers), along with service logs, media, documents and preventative maintenance linkage.

Stage-Gated Project Management

Projects follow a structured eight-stage lifecycle from enquiry through to completion sign-off. Each stage carries its own task checklist, notes and file storage. Approval-required stages remain locked until the preceding stage is validated, ensuring no work proceeds without the necessary sign-offs from management or the customer.

Clearance and Classification Management

Customers can be marked as classified, restricting access to their sites, jobs, notes and linked data to authorised users only. Engineers without clearance see the customer name and a clearance-required label. A request and approval workflow allows engineers to apply for access, with Admin and Management users able to grant or revoke clearance at any time.

Compliance Document Generation

Compliance certificates and documents are generated from predefined templates, with site, job, customer and asset data auto-filled from the platform. Generated documents are linked to their originating context, stored per customer and site, and available to download or print as PDF, with a compliance report available for audit purposes.

How Does It Work?

1

Platform Setup

The company profile, VAT rates, callout fee bands, labour rates and default system settings are configured. Admin users are created and assigned roles, and the equipment and parts libraries are populated ready for operational use.

2

Customer & Site Creation

Customers are added with full contact, account manager and classification details. One or more sites are created under each customer, with address information, floor plans, photos and contacts, giving the platform a complete geographic picture of the operation.

3

Project or Job Planning

For larger deployments, a project is created and progressed through its stage-gated lifecycle. For service and maintenance work, jobs are created directly with tasks, parts, equipment, scheduling and engineer assignment, with the cost estimation engine calculating the full job value automatically.

4

Field Execution

Engineers work through job tasks, logging completion notes, photos, videos and audio as they go. Site visits are recorded with arrival and departure times and actual labour and parts used. Installed equipment is converted into tracked asset records directly from the job card.

5

Compliance & Completion

Compliance documents are generated from templates auto-filled with job, site and asset data and exported as PDF. For projects, the completion stage requires customer sign-off and handover documentation before the project is marked as closed.

Outcomes & Impact

Unified Security Operations

Platform

Customers, sites, jobs, equipment, parts, assets, projects and compliance are connected in one platform. Teams manage the full operational workflow without switching between tools or maintaining parallel records across systems.

Controlled Data Security

Access

Role-based access control and clearance management ensure that sensitive customer records, classified sites and restricted operational data are visible only to authorised personnel, with a full audit trail of every action across the platform.

Accurate Job Costing

Commercial

The automated cost estimation engine applies the correct VAT rates and callout fees per job type, replacing manual calculations and reducing the risk of pricing errors across labour, parts and callout charges.

Technologies Used to Develop the Solution

Frontend

React

Web management panel with rolebased module visibility, tile-based dashboard navigation and responsive operational interfaces.

Backend

Node.js

API layer managing customers, sites, jobs, equipment, parts, assets, projects, compliance documents, users and audit logging.

Database

PostgreSQL

Customer, site, job, asset, equipment, parts, project, compliance and audit log data with relational integrity across all linked entities.

Authentication

Email / Password + 2FA

Secure login with two-factor authentication via Google or Microsoft Authenticator and a self-service 2FA reset workflow managed by Senior Management.

Access Control

RBAC and Clearance Management

Predefined roles controlling module visibility and data access, with a clearance layer restricting classified customer data to authorised users via a request and approval workflow.

Documents

PDF Generation

Compliance documents generated from predefined templates, auto-filled with platform data and exported as downloadable and printable PDFs.

Mapping

Geographic Map Integration

Customer location map view with geographic search, country-based filtering and location-based exploration across international deployments.

Data Exchange

Excel Import / Export

Parts catalogue bulk import and export via Excel, supporting large catalogue migrations and ongoing parts management outside the platform interface.

Security

Encryption and Audit Logging

Data encrypted at rest and in transit, with comprehensive audit logs recording all user actions across every module for traceability and compliance.

Here's What Our Client Has to Say

> Having everything in one system (customers, sites, jobs, equipment, assets and compliance) has made a real difference to how we operate. The job costing engine handles the VAT calculations automatically, which removes a lot of manual work and reduces the risk of errors. The clearance management gives us confidence that sensitive customer data is only accessible to the right people.

Killian Darcy

Founder, Pulse Security

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REVIEWED ON Clutch ★★★★★ 27 REVIEWS

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